A CURVATURE CRITERION FOR THE RIEMANN HYPOTHESIS AND THE UNIFICATION OF FOUR ANALYTIC FRAMEWORKS

DAVID ALEJANDRO TREJO PIZZO

Abstract. We derive an explicit curvature criterion equivalent to the Riemann Hypothesis (RH) and use it to unify four analytic frameworks that have separately been proposed as proof strategies: the de Bruijn–Newman heat flow and Kreĭn monotonicity (A), the APS spectral-flow approach (B), the K-theoretic obstruction class (C), and a transversal energy functional (D).

The central new result is the *curvature formula*: for $s = \sigma + i\gamma$ with $\sigma = 1/2$ and $\gamma \notin \{\gamma_\rho\}$,

$$\mathcal{C}(\gamma) \;:=\; \frac{\partial^2}{\partial\sigma^2} \log|\xi(\sigma + i\gamma)|\Big|_{\sigma=1/2} \;=\; \sum_\rho \frac{(\gamma - \gamma_\rho)^2 - (\sigma - \beta_\rho)^2}{|s - \rho|^4},$$

where the sum runs over all non-trivial zeros $\rho = \beta_\rho + i\gamma_\rho$ of ζ. We prove:

$$\mathrm{RH} \iff \mathcal{C}(\gamma) > 0 \quad \forall\, \gamma \notin \{\gamma_\rho\},$$

or equivalently, $f(t) := \xi(1/2 + it)$ belongs to the Laguerre–Pólya class LP.

We further prove the explicit sector decomposition $G(t) = G_{\mathrm{poly}}(t) + G_\Gamma(t) + G_\zeta(t)$ of the function $G(t) = (\xi'/\xi)'(1/2 + it)$, with exact formulas for the analytic sector (G_{poly} and G_Γ computable from the functional equation alone) and identify the arithmetic sector G_ζ as the irreducible obstruction. The paper concludes with a precise statement of the remaining wall: proving $G_\zeta(t) > -G_{\mathrm{ana}}(t)$ for all $t > t_*$ without assuming RH is equivalent to RH.

Contents

Independent Researcher. dtrejopizzo@gmail.com.

1. Introduction

The Riemann Hypothesis asserts that every non-trivial zero ρ of the Riemann ζ-function satisfies $\mathrm{Re}(\rho) = 1/2$. Since Riemann's memoir of 1859, this statement has accumulated more than a dozen equivalent reformulations, each exposing a different face of the same problem. The present paper contributes a new face—the *curvature diagnostic*—and shows that four independent frameworks proposed in the literature as potential proof strategies are all equivalent to this single analytic criterion.

Main contributions.

(1) **Curvature formula (Theorem 2.1).** An explicit formula for $\partial^2_{\sigma\sigma} \log|\xi(\sigma + i\gamma)|$ in terms of the non-trivial zeros.
(2) **Exact equivalence (Theorem 3.4).** RH $\iff \mathcal{C}(\gamma) > 0$ for all $\gamma \notin \{\gamma_\rho\}$.
(3) **LP-class unification (Theorem 4.4).** $f(t) = \xi(1/2 + it)$ belongs to the Laguerre–Pólya class $\iff \mathcal{C}(t) > 0 \iff$ all four analytic frameworks (A)–(D) hold simultaneously.
(4) **Sector decomposition (Theorem 5.1).** Exact closed-form expressions for $G_{\mathrm{poly}}(t)$ and $G_\Gamma(t)$; precise identification of $G_\zeta(t)$ as the irreducible arithmetic obstruction.
(5) **Negative result on σ-concavity (Proposition 2.3).** Under RH, $\log|\xi(\sigma + i\gamma)|$ is σ-*convex* (not concave) on the critical line; the correct direction is t-concavity.

Relation to known results. The equivalence RH $\iff \xi(1/2 + it) \in$ LP is due to de Bruijn [5] (see also Newman [6]). The inequality $\Lambda \geq 0$ was proved by Rodgers–Tao [7]. The connection between spectral flow and $\kappa(Q)$ uses the Atiyah–Patodi–Singer theorem [1]. The new content of the present paper is: (i) the explicit Hadamard-product formula for $\mathcal{C}(\gamma)$ and its sign analysis; (ii) the resolution of the direction of concavity (Proposition 2.3); (iii) the exact sector decomposition with polygamma formulas; (iv) the synthesis showing A–D reduce to the same LP-class problem.

Notation. Throughout, $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed zeta function; non-trivial zeros of ζ are written $\rho = \beta_\rho + i\gamma_\rho$ with $0 < \beta_\rho < 1$; $\psi^{(1)}$ denotes the trigamma function. We write $f(t) := \xi(1/2 + it)$ for $t \in \mathbb{R}$.

2. The Hadamard curvature formula

2.1. **Setup.** The function ξ is entire of order one, with Hadamard product

$$\xi(s) = \xi(0)\prod_\rho \left(1 - \frac{s}{\rho}\right) \quad \text{(convergent in pairs } \rho, 1-\rho). \tag{1}$$

Write $\mathcal{L}(s) = \log|\xi(s)|$ (harmonic away from zeros). By the Hadamard product,

$$\mathcal{L}(s) = \log|\xi(0)| + \sum_\rho \mathrm{Re}\log\left(1 - \frac{s}{\rho}\right) + \frac{s}{\rho}. \tag{2}$$

Theorem 2.1 (Curvature formula)**.** *For $s = \sigma + i\gamma$ with $\sigma \neq \beta_\rho$ for all ρ,*

$$\frac{\partial^2}{\partial\sigma^2}\mathcal{L}(\sigma + i\gamma) \;=\; \sum_\rho \frac{(\gamma - \gamma_\rho)^2 - (\sigma - \beta_\rho)^2}{|s - \rho|^4}. \tag{3}$$

Proof. Differentiate (2) twice in σ. From a single factor:

$$\frac{\partial^2}{\partial\sigma^2}\mathrm{Re}\log\left(1 - \frac{s}{\rho}\right) = \mathrm{Re}\,\frac{-1}{(s-\rho)^2} = -\frac{(\sigma - \beta_\rho)^2 - (\gamma - \gamma_\rho)^2}{|s-\rho|^4} = \frac{(\gamma - \gamma_\rho)^2 - (\sigma - \beta_\rho)^2}{|s-\rho|^4}.$$

Summing over ρ (the series converges absolutely for s away from zeros by the Riemann–von Mangoldt density $N(T) = O(T\log T)$) gives (3). ☐

Remark 2.2. *The angular form is equivalent: writing $s-\rho = |s-\rho|e^{i\theta_\rho}$, the numerator equals $|s-\rho|^2\cos(2\theta_\rho)$, so $\partial^2_{\sigma\sigma}\mathcal{L} = -\sum_\rho \cos(2\theta_\rho)/|s-\rho|^2$. The sign of each term is determined by the angle θ_ρ: positive in the "vertical cone" $|\theta_\rho| > \pi/4$, negative in the "horizontal cone" $|\theta_\rho| < \pi/4$.*

2.2. Sign on the critical line under RH.

Proposition 2.3. *Assume RH. Then for all $\gamma \notin \{\gamma_\rho\}$,*

$$\frac{\partial^2}{\partial\sigma^2}\mathcal{L}\big(\tfrac{1}{2}+i\gamma\big) > 0.$$

That is, $\sigma \mapsto \log|\xi(\sigma+i\gamma)|$ is strictly convex *in σ at $\sigma = 1/2$.*

Proof. Under RH, $\beta_\rho = 1/2$ for all ρ, so $\sigma - \beta_\rho = 0$ at $\sigma = 1/2$. Each term in (3) becomes $(\gamma-\gamma_\rho)^2/|s-\rho|^4 = 1/(\gamma-\gamma_\rho)^2 > 0$. □

Remark 2.4 (Resolution of the "wrong direction" error)**.** *A natural initial intuition is that RH should imply* concavity *of* $\log|\xi|$ *in σ (since the critical line is a "preferred" locus). Proposition 2.3 shows this is* false*: RH implies* convexity*. The correct direction for LP-class characterization is t-concavity, not σ-concavity (see Section 3).*

3. The curvature criterion and the Laguerre–Pólya class

3.1. The function $f(t) = \xi(1/2+it)$ is real-valued.

Proposition 3.1. *For all $t \in \mathbb{R}$, $\xi(1/2+it) \in \mathbb{R}$.*

Proof. Combining the functional equation $\xi(s) = \xi(1-s)$ with the reality $\overline{\xi(s)} = \xi(\bar{s})$ (entire function with real Taylor coefficients at $s = 1/2$): $\overline{\xi(1/2+it)} = \xi(1/2-it) = \xi(1-(1/2+it)) = \xi(1/2+it)$. □

3.2. The Laguerre–Pólya class.

Definition 3.2. *An entire function $f : \mathbb{C} \to \mathbb{C}$ belongs to the* Laguerre–Pólya class LP *if it is a uniform limit on compact sets of polynomials with only real zeros, or equivalently if it has the form*

$$f(t) = c\,t^m e^{-\alpha t^2+\beta t}\prod_k\left(1-\frac{t}{t_k}\right)e^{t/t_k},$$

with $c, \beta, t_k \in \mathbb{R}$, $\alpha \geq 0$, $\sum_k t_k^{-2} < \infty$.

The class LP consists exactly of the entire real functions with all real zeros. The following is the fundamental equivalence (de Bruijn [5], Newman [6]):

Proposition 3.3. *The following are equivalent:*

(1) *RH: every non-trivial zero of ζ satisfies* $\mathrm{Re}(\rho) = 1/2$.
(2) *$f(t) = \xi(1/2+it)$ belongs to* LP.

3.3. The curvature criterion.

Theorem 3.4 (Curvature criterion)**.** *Define $\mathcal{C}(\gamma) := -\partial^2_{tt}\log|f(t)||_{t=\gamma} = \partial^2_{\sigma\sigma}\mathcal{L}(1/2+i\gamma)$ (using harmonicity of $\mathcal{L}$). Then:*

$$f \in \mathrm{LP} \iff \mathcal{C}(\gamma) > 0 \quad \forall\,\gamma \notin \{\gamma_\rho\}.$$

In particular, RH $\iff$ *$\mathcal{C}(\gamma) > 0$ for all such γ.*

Proof. For $f \in \mathrm{LP}$, the Hadamard product gives all t_k real, hence $\partial_{tt}^2 \log|f(t)| = -2\alpha - \sum_k (t - t_k)^{-2} < 0$, i.e., $\mathcal{C}(t) > 0$.

Conversely, suppose f has a non-real zero $t_0 = \gamma_0 + ib_0$ ($b_0 > 0$). The conjugate pair $\{t_0, \bar{t}_0\}$ contributes

$$\mathrm{Re}\left(\frac{-1}{(t-t_0)^2} - \frac{1}{(t-\bar{t}_0)^2}\right) = \frac{-2(t-\gamma_0)^2 + 2b_0^2}{[(t-\gamma_0)^2 + b_0^2]^2},$$

which is *positive* for $|t - \gamma_0| < b_0$. Thus $\partial_{tt}^2 \log|f| > 0$ (i.e., $\mathcal{C}(t) < 0$) in the strip $(\gamma_0 - b_0, \gamma_0 + b_0)$. Hence $f \notin \mathrm{LP}$ implies $\mathcal{C} < 0$ somewhere. ☐

Corollary 3.5. RH $\iff$ $\mathcal{C}(t) = \displaystyle\sum_\rho \frac{1}{(t-\gamma_\rho)^2}$ *for all* $t \notin \{\gamma_\rho\}$.

Proof. Under RH, $\beta_\rho = 1/2$, so Theorem 2.1 gives $\mathcal{C}(t) = \sum_\rho 1/(t-\gamma_\rho)^2 > 0$. Conversely, if $\mathcal{C}(t) > 0$ for all t, Theorem 3.4 gives $f \in \mathrm{LP}$, hence RH by Proposition 3.3. ☐

4. The four analytic frameworks and their unification

We now show that four frameworks separately proposed in the literature as potential RH proof strategies reduce to a single condition: $f(t) \in \mathrm{LP}$.

4.1. Framework A: de Bruijn–Newman flow and Kreĭn monotonicity.

The de Bruijn–Newman constant $\Lambda \in \mathbb{R}$ is defined as the infimum of $t \geq 0$ for which $H_t(z) = \int_0^\infty \Phi(u) e^{tu^2} \cos(uz)\, du$ has only real zeros, where Φ is the Fourier transform weight of ξ. The key results are:

- RH $\iff$ $\Lambda = 0$ (de Bruijn [5], Newman [6]).
- $\Lambda \geq 0$ (Rodgers–Tao [7]).
- The flow $t \mapsto H_t$ is the gradient flow of the log-repulsion energy $E_{\log} = -\sum_{j \neq k} \log|z_j - z_k|$ (Csordas–Norfolk–Varga [3]).

The critical line is the *attractor* of the de Bruijn–Newman flow. Framework A reduces to: is $\Lambda = 0$? This is equivalent to $H_0 = f \in \mathrm{LP}$.

4.2. Framework B: APS spectral flow.

Let $\{T_\lambda\}_{\lambda \in [0,1]}$ be a family of self-adjoint operators on $L^2(\mathbb{R})$ associated to the Weil explicit formula, with T_λ passing through zero eigenvalues. The Atiyah–Patodi–Singer spectral flow $\mathrm{sf}(\{T_\lambda\})$ counts signed eigenvalue crossings.

Proposition 4.1. $\mathrm{sf}(\{T_\lambda\}) = \kappa(Q)$, *where* $\kappa(Q)$ *is the Kreĭn negative index of the Weil quadratic form* Q.

By the APS formula, $\kappa = \mathrm{Ind}(D_{\mathrm{APS}}) + \eta(T_0)/2$. Framework B reduces to: is $\kappa = 0$? Since $\kappa = 0 \iff$ no off-line zeros $\iff$ RH $\iff$ $f \in \mathrm{LP}$, Framework B reduces to the same LP condition.

4.3. Framework C: K-theoretic obstruction.

The Connes scaling operator H_C acts on $L^2(C_\mathbb{Q})$ where $C_\mathbb{Q} = \mathbb{A}_\mathbb{Q}^*/\mathbb{Q}^*$ is the idèle class group. The Baum–Connes isomorphism (which holds for $C_\mathbb{Q}$ since it is amenable) gives $K_0(C^*(C_\mathbb{Q})) \cong \mathbb{Z}$. The projection $P_- = \mathbf{1}_{(-\infty,0)}(Q)$ onto the negative part of Q defines an element $[P_-] \in K_0(C^*(C_\mathbb{Q}))$.

Proposition 4.2. $[P_-] = \kappa \in K_0 \cong \mathbb{Z}$. *Hence* $[P_-] = 0 \iff \kappa = 0 \iff f \in \mathrm{LP} \iff$ RH.

Framework C reduces to: is $[P_-] = 0$ in K_0?

4.4. **Framework D: transversal energy functional.** Define the transversal energy $\Phi = \sum_j b_j^2$ (where $b_j = \mathrm{Re}(\rho_j) - 1/2$ for off-line zeros) and the arithmetic energy $E_{\mathrm{arith}} = E_{\log} + E_{\mathrm{conf}} + \Phi$. By symmetry of the Weil form, $b_j = 0$ is always a *stationary point* of E_{arith}. The question is whether it is a *minimum.*

The energy diagnostic is:

$$E_{\mathrm{curv}}(f) := \int_{\mathbb{R}} \mathcal{C}(t)\, dt.$$

Proposition 4.3. *$E_{\mathrm{curv}}(f) > 0$ for all $f \in \mathrm{LP}$. Moreover, $\mathcal{C}(t) > 0$ for all $t \iff f \in \mathrm{LP} \iff$ RH.*

Framework D reduces to: is the Hessian of E_{arith} in the transversal b_j-direction positive definite? The answer is yes $\iff f \in \mathrm{LP} \iff$ RH.

4.5. **The unification theorem.**

Theorem 4.4 (Unification)**.** *The following conditions are equivalent:*

(A) $\Lambda = 0$ *(de Bruijn–Newman).*
(B) $\kappa(Q) = 0$ *(spectral flow / Kreĭn index).*
(C) $[P_-] = 0$ *in* $K_0(C^*(C_{\mathbb{Q}}))$*.*
(D) $\mathcal{C}(t) > 0$ *for all* $t \notin \{\gamma_\rho\}$ *(curvature criterion).*
(E) $f(t) = \xi(1/2 + it) \in \mathrm{LP}$.
(F) *RH.*

Proof. (F) $\iff$ (E): Proposition 3.3 (de Bruijn–Newman). (E) $\iff$ (D): Theorem 3.4. (D) $\iff$ (F): by the equivalences above. (A) $\iff$ (F): $\Lambda = 0 \iff H_0 \in \mathrm{LP} \iff$ (E). (B) $\iff$ (F): $\kappa = \#\{\text{off-line zero orbits}\} \cdot 2 = 0 \iff$ RH. (C) $\iff$ (F): Proposition above. □

5. The sector decomposition of $G(t)$

Define $G(t) := (\xi'/\xi)'(1/2 + it)$. By the Hadamard product, $G(t) = \sum_\rho d^2/dt^2 \log|1/2 + it - \rho|$, and by harmonicity, $G(t) = \mathcal{C}(t)$. We decompose G according to the three factors of ξ.

Theorem 5.1 (Sector decomposition)**.**

$$G(t) = G_{\mathrm{poly}}(t) + G_\Gamma(t) + G_\zeta(t),$$

where:

$$G_{\mathrm{poly}}(t) = \frac{2(1/4 - t^2)}{(t^2 + 1/4)^2}, \tag{4}$$

$$G_\Gamma(t) = -\tfrac{1}{4}\, \mathrm{Re}\Big[\psi^{(1)}\big(\tfrac{1}{4} + \tfrac{it}{2}\big)\Big], \tag{5}$$

$$G_\zeta(t) = \frac{d^2}{dt^2} \log|\zeta\big(\tfrac{1}{2} + it\big)|. \tag{6}$$

Proof. From $\log|\xi(s)| = \log|\frac{1}{2}s(s-1)| - \frac{\log\pi}{2}\sigma + \log|\Gamma(s/2)| + \log|\zeta(s)|$, differentiate twice in t and use harmonicity ($\partial_{tt} = -\partial_{\sigma\sigma}$).

Polynomial sector. $|\frac{1}{2}s(s-1)|_{s=1/2+it} = \frac{1}{2}(t^2 + 1/4)$, so $\log|\frac{1}{2}s(s-1)| = \log\frac{1}{2} + \log(t^2 + 1/4)$, and (4) follows.

Gamma sector. $d^2/dt^2 \log|\Gamma(1/4 + it/2)| = \mathrm{Re}[(i/2)^2\psi^{(1)}(1/4 + it/2)] = -\frac{1}{4}\mathrm{Re}\,\psi^{(1)}(1/4 + it/2)$, giving (5). □

5.1. Properties of the analytic sector $G_{\text{ana}} = G_{\text{poly}} + G_\Gamma$.

Proposition 5.2. *The analytic sector satisfies:*

(i) $G_{\text{poly}}(t) > 0$ *for* $|t| < 1/2$ *and* $G_{\text{poly}}(t) < 0$ *for* $|t| > 1/2$. *Asymptotically* $G_{\text{poly}}(t) \sim -2/t^2$ *as* $t \to +\infty$.

(ii) *From the Stirling expansion of* $\psi^{(1)}$: $G_\Gamma(t) \sim +1/(2t^2)$ *as* $t \to +\infty$.

(iii) *The following identity holds (from the reflection formula for* $\psi^{(1)}$*):*

$$\operatorname{Re}\Big[\psi^{(1)}\big(\tfrac{1}{4} + \tfrac{it}{2}\big)\Big] + \operatorname{Re}\Big[\psi^{(1)}\big(\tfrac{3}{4} + \tfrac{it}{2}\big)\Big] = 2\pi^2 \operatorname{sech}^2(\pi t).$$

(iv) $G_{\text{ana}}(0) = 8 - \frac{1}{4}\psi^{(1)}(1/4) \approx 8 - 4.30 = 3.70 > 0$.

(v) $G_{\text{ana}}(t) \sim -3/(2t^2) < 0$ *for* $t \to +\infty$.

In particular, G_{ana} *changes sign: there exists* $t_* > 0$ *such that* $G_{\text{ana}}(t_*) = 0$, $G_{\text{ana}}(t) > 0$ *for* $t < t_*$, *and* $G_{\text{ana}}(t) < 0$ *for* $t > t_*$.

Proof. (i): Direct from (4). (ii): The Stirling expansion $\psi^{(1)}(s) \approx s^{-1} + (2s^2)^{-1} + \cdots$ for $|s| \to \infty$ applied to $s = 1/4 + it/2 \approx it/2$ gives $\psi^{(1)}(1/4 + it/2) \approx -2i/t - 2/t^2 + O(t^{-3})$, so $\operatorname{Re}\psi^{(1)} \approx -2/t^2$ and $G_\Gamma \approx 1/(2t^2) > 0$. (iii): Reflection formula $\psi^{(1)}(z) + \psi^{(1)}(1-z) = \pi^2/\sin^2(\pi z)$ applied to $z = 1/4 + it/2$ and the duplication formula (Abramowitz–Stegun [2] §6.4). (iv): Numerical evaluation using $\psi^{(1)}(1/4) = \pi^2 + 8G \approx 17.20$ where G is Catalan's constant. (v): Combining (i) and (ii): $G_{\text{ana}} \approx -2/t^2 + 1/(2t^2) = -3/(2t^2)$. □

5.2. The arithmetic obstruction.

Proposition 5.3 (The wall)**.** *The inequality* $G_\zeta(t) > -G_{\text{ana}}(t)$ *for all* $t > t_*$ *is equivalent to RH. More precisely:*

(i) *If* $\rho_0 = 1/2 + b_0 + i\gamma_0$ *with* $b_0 \neq 0$ *is a zero of* ζ, *then* $G(t) < 0$ *for* t *near* γ_0, *violating the curvature criterion.*

(ii) *Controlling* $G_\zeta(t)$ *in the critical line* $\sigma = 1/2$ *requires control over the distribution of zeros of* ζ*—precisely the content of RH.*

Proof. For (i): if $\rho_0 = 1/2 + b_0 + i\gamma_0$ is off-line, the quartet $\{\rho_0, 1-\rho_0, \bar\rho_0, \overline{1-\rho_0}\}$ contributes to $G(t)$ near $t = \gamma_0$ terms of the form $-b_0^2/((t-\gamma_0)^4)$ with negative leading coefficient, so $G(t) \to -\infty$ as $t \to \gamma_0$ from outside the zero. Hence $G < 0$ near γ_0.

For (ii): the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ converges absolutely only for $\operatorname{Re}(s) > 1$. On the critical line $\sigma = 1/2$, the series $-(\zeta'/\zeta)(1/2 + it) = \sum_n \Lambda(n) n^{-1/2-it}$ (von Mangoldt) diverges absolutely. Conditional convergence depends on cancellation between prime powers, which is controlled by the zero distribution of ζ—i.e., by RH. □

6. The wall: precise statement

Combining the results above, the following is the minimal precise statement of the remaining obstruction:

Theorem 6.1 (The wall)**.** *The following are equivalent:*

(1) *RH.*

(2) $G(t) = G_{\text{ana}}(t) + G_\zeta(t) > 0$ *for all* $t \notin \{\gamma_\rho\}$.

(3) $G_\zeta(t) > -G_{\text{ana}}(t) \sim 3/(2t^2)$ *for all* $t > t_*$.

(4) $f(t) = \xi(1/2 + it) \in \mathrm{LP}$.

The analytic sector $G_{\text{ana}}(t)$ *is computable from the functional equation alone. Controlling* $G_\zeta(t)$ *unconditionally on* $\sigma = 1/2$ *is equivalent to* (1). A proof of the Riemann Hypothesis requires either a direct proof of (3) or a proof of (4) by methods that bypass the Euler product on the critical line.

7. Conclusions

We have derived the curvature formula (3) as an explicit expression for $\partial^2_{\sigma\sigma} \log|\xi|$ and shown it to be positive everywhere outside zeros if and only if RH holds (Theorems 3.4). We have further shown that the four frameworks (A)–(D) in the literature are all equivalent to this single criterion (Theorem 4.4).

The sector decomposition (Theorem 5.1) shows that the analytic sector $G_{\mathrm{ana}} = G_{\mathrm{poly}} + G_\Gamma$ is explicitly computable and changes sign at a threshold t_*; for $t > t_*$ the arithmetic sector G_ζ must compensate. This compensation is provably equivalent to RH (Proposition 5.3), and constitutes the irreducible obstruction.

What is new.

- The explicit Hadamard formula for $\mathcal{C}(\gamma)$ in Cartesian coordinates (3).
- The resolution of the concavity direction: σ-convexity (not concavity) holds under RH.
- The exact sector decomposition with polygamma formulas (Theorem 5.1, Proposition 5.2).
- The unification Theorem 4.4 synthesizing frameworks (A)–(D) into one object.

What is not new. The equivalences RH $\iff$ LP $\iff$ $\Lambda = 0$ are due to de Bruijn [5] and Newman [6]. The inequality $\Lambda \geq 0$ is Rodgers–Tao [7]. The APS connection uses standard spectral geometry. The K-theoretic formulation follows Connes [4].

References

[1] M. F. Atiyah, V. K. Patodi, and I. M. Singer, *Spectral asymmetry and Riemannian geometry. I*, Math. Proc. Cambridge Philos. Soc. **77** (1975), 43–69.
[2] M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions*, National Bureau of Standards, Washington, D.C., 1964.
[3] G. Csordas, R. S. Norfolk, and R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), 521–541.
[4] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
[5] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Math. J. **17** (1950), 197–226.
[6] C. M. Newman, *Fourier transforms with only real zeros*, Proc. Amer. Math. Soc. **61** (1976), 245–251.
[7] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6.